

Week 7 (first part)

Linear transformations between vector spaces:

Definition 4.26: Let V, W be two vector spaces. A function $T : V \rightarrow W$ is called a *linear transformation (between vector spaces)* if the following linearity axiom holds for all $\mathbf{x}_1, \mathbf{x}_2 \in V$ and all $\lambda_1, \lambda_2 \in \mathbb{R}$.

$$\underbrace{T(\underbrace{\lambda_1 \mathbf{x}_1 + \lambda_2 \mathbf{x}_2}_{\substack{\in V, \text{Lemma 4.16}}})}_{\in W} = \underbrace{\lambda_1 \underbrace{T(\mathbf{x}_1)}_{\in W}}_{\in W, \text{Lemma 4.16}} + \underbrace{\lambda_2 \underbrace{T(\mathbf{x}_2)}_{\in W}}_{\in W, \text{Lemma 4.16}}.$$

So far: $V = \mathbb{R}^n, W = \mathbb{R}^m$.

Different example: Matrix “flattening”:

$$V = \mathbb{R}^{2 \times 2}, \quad W = \mathbb{R}^4, \quad T : \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}.$$

This is *bijective* (undoable).

Lemma 4.27: Let $T : V \rightarrow W$ be a bijective linear transformation, $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_\ell\} \subseteq V$, $T(B) = \{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_\ell)\} \subseteq W$. Then

- $|T(B)| = |B|$.
- B is a basis of $V \Leftrightarrow T(B)$ is a basis of W . We therefore also have $\dim(V) = \dim(W)$.

Definition 4.28: If there is a bijective linear transformation $T : V \rightarrow W$, then V and W are called *isomorphic*, and T is called an *isomorphism* between V and W .

Isomorphism examples:

- $V = \mathbb{R}^{m \times n}, W = \mathbb{R}^{nm}$, T the “flattening” transformation

- $V = \text{set of degree-2 polynomials}, W = \mathbb{R}^3$, $T : p_0 + p_1x + p_2x^2 \mapsto \begin{pmatrix} p_0 \\ p_1 \\ p_2 \end{pmatrix}$

Computing (bases of) the three fundamental subspaces (Section 4.3)

Column space $C(A)$, row space $R(A)$ and nullspace $N(A)$ of a matrix A .

Based on $A \rightarrow R$ (Gauss-Jordan):

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{bmatrix} \rightarrow R = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Column space:

Theorem 4.31: If R in $\text{RREF}(j_1, j_2, \dots, j_r)$, then A has its independent columns at indices $j_1, j_2, \dots, j_r$, and these columns form a **basis** of the column space $\mathbf{C}(A)$. In particular,

$$\dim(\mathbf{C}(A)) = \text{rank}(A) = r.$$

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{bmatrix} \rightarrow R = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$j_1 \quad j_2$

Proof. Independent columns of A : basis of $\mathbf{C}(A)$ (Lemma 4.19), and R reveals their indices $j_1, j_2, \dots, j_r$ (Theorem 3.18). Thus $\dim(\mathbf{C}(A)) = r$ (Definition 4.25) and $\text{rank}(A) = r$ (Definition 2.10). $\square$

Row space:

Theorem 4.32: If R in $\text{RREF}(j_1, j_2, \dots, j_r)$, the first r rows of R form a basis of the row space $\mathbf{R}(A)$. In particular,

$$\dim(\mathbf{R}(A)) = r.$$

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{bmatrix} \rightarrow R = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Proof. $R = MA$ with M invertible (Theorem 3.17) $\Rightarrow R$ and A have the same row space (Lemma 3.5). The r nonzero rows of R already span the row space, and they are linearly independent (private nonzeros at the downward steps). So the first r rows are a basis and $\dim(\mathbf{R}(A)) = r$ (Definition 4.25). $\square$

Theorem 4.33: Let A be an $m \times n$ matrix. Then $\text{rank}(A) = \text{rank}(A^\top)$.

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{bmatrix} \quad A^\top = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 2 \\ 3 & 4 & 5 \end{bmatrix}$$

$\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 = 2\mathbf{v}_2 - \mathbf{v}_1$

Proof.

- $\text{rank}(A) = \dim(\mathbf{C}(A)) = r$ (Theorem 4.31)
- $\text{rank}(A^\top) = \dim(\mathbf{C}(A^\top)) \underbrace{=}_{\mathbf{R}(A) = \mathbf{C}(A^\top)} \dim(\mathbf{R}(A)) = r$ (Theorem 4.32)

$\square$

Corollary 4.34: Let A be an $m \times n$ matrix of rank r . Then $r \leq \min(n, m)$.

Proof. $r = \text{rank}(A) \leq n$ (number of columns), $r = \text{rank}(A^\top) \leq m$ (number of rows). $\square$

Nullspace:

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{bmatrix} \rightarrow R = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow R' = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

Lemma 3.3: $\mathbf{N}(A) = \mathbf{N}(R)(= \mathbf{N}(R'))$.

Dimension / basis by example: $\mathbf{x} \in \mathbf{N}(R') \Leftrightarrow R'\mathbf{x} = \mathbf{0}$, and this can be written as

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \underbrace{\begin{pmatrix} x_1 \\ x_3 \end{pmatrix}}_{\in \mathbb{R}^r} + \begin{bmatrix} 2 & 3 \\ 0 & -2 \end{bmatrix} \underbrace{\begin{pmatrix} x_2 \\ x_4 \end{pmatrix}}_{\in \mathbb{R}^{n-r}} = \mathbf{0}. \quad \text{Or:} \quad \begin{pmatrix} x_1 \\ x_3 \end{pmatrix} = - \begin{bmatrix} 2 & 3 \\ 0 & -2 \end{bmatrix} \begin{pmatrix} x_2 \\ x_4 \end{pmatrix}. \quad (*)$$

$\Rightarrow \mathbf{N}(R')$ is isomorphic to $\mathbb{R}^{n-r}$ and therefore has dimension $n - r$ (Lemma 4.27):

$$T : \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}}_{\in \mathbf{N}(R')} \mapsto \underbrace{\begin{pmatrix} x_2 \\ x_4 \end{pmatrix}}_{\in \mathbb{R}^{n-r}}$$

is a bijective linear transformation:

“for every possible output $\begin{pmatrix} x_2 \\ x_4 \end{pmatrix} \in \mathbb{R}^{n-r}$, exactly one input $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbf{N}(R')$ leads to it.”

We get the missing input values x_1, x_3 from (*).

Basis of $\mathbf{N}(A) = \mathbf{N}(R')$: inputs that lead to outputs $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_{n-r}$ (Theorem 4.36).

Summary:

$\mathbf{C}(A)$ (dim = r)	$\mathbf{N}(A)$ (dim = $n - r$)	$\mathbf{R}(A)$ (dim = r)
Bases: $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$	$\underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{x_2, x_4} \quad \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{x_2, x_4}$ $\underbrace{\begin{pmatrix} -2 \\ 0 \end{pmatrix}}_{x_1, x_3} \quad \underbrace{\begin{pmatrix} -3 \\ 2 \end{pmatrix}}_{x_2, x_3}$	$\begin{pmatrix} 1 \\ 2 \\ 0 \\ 3 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 1 \\ -2 \end{pmatrix}$
$\left[\begin{array}{cc} 1 & 0 \\ 2 & 1 \\ 3 & 2 \end{array} \right]_C \leftarrow$	$\left[\begin{array}{cccc} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{array} \right]_{A=CR'} \rightarrow$	$\left[\begin{array}{cccc} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{array} \right]_{R \text{ in RREF}(1,3)} \rightarrow$
$\left[\begin{array}{cc} 1 & 0 \\ 2 & 1 \\ 3 & 2 \end{array} \right]_C$	$\left[\begin{array}{cccc} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{array} \right]_{A=CR'}$	$\left[\begin{array}{cccc} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{array} \right]_{R' \text{ in RREF}(1,3)}$
Theorem 4.31	Theorem 4.36	Theorem 4.32

All solutions of $Ax = b$ (Section 4.4)

Definition 4.37: Let A be an $m \times n$ matrix and $b \in \mathbb{R}^m$. The set

$$\text{Sol}(A, b) := \{x \in \mathbb{R}^n : Ax = b\} \subseteq \mathbb{R}^n$$

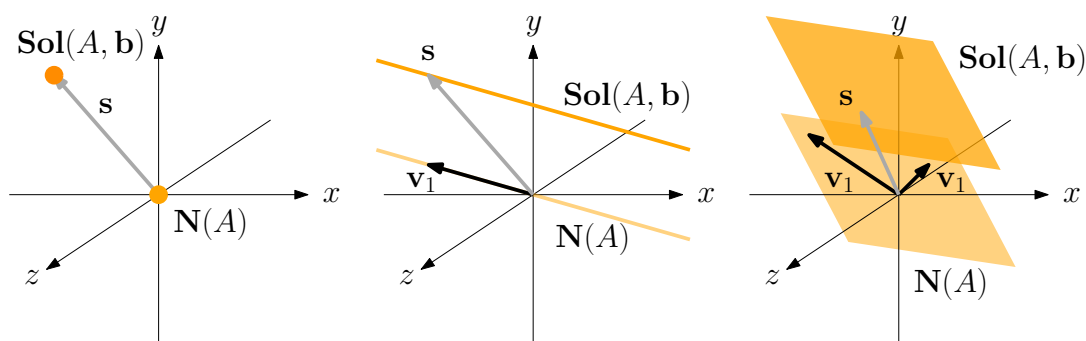
is the *solution space* of $Ax = b$.

Not a subspace if $b \neq 0$, because then $0 \notin \text{Sol}(A, b)$ (see Lemma 4.9).

$$\text{Sol}(A, 0) = N(A).$$

$\text{Sol}(A, b)$ = “shifted nullspace”:

Theorem 4.38: Let s be some solution of $Ax = b$. Then $\text{Sol}(A, b) = \{s + x : x \in N(A)\}$.



Proof.

(i) for every $y \in \text{Sol}(A, b)$, there is $x \in N(A)$ with $y = s + x$. To see this, write

$$y = s + \underbrace{(y - s)}_x.$$

Then $Ax = A(y - s) = Ay - As = b - b = 0 \Rightarrow x \in N(A)$.

(ii) for every vector $y = s + x$ with $x \in N(A)$ we have $y \in \text{Sol}(A, b)$. To see this:

$$Ay = A(s + x) = As + Ax = b + 0 = b.$$

□

Theorem 4.39: Let A be an $m \times n$ matrix of rank r . If $Ax = b$ has a solution, then $\text{Sol}(A, b)$ has dimension $n - r$, where $\dim(\text{Sol}(A, b)) := \dim(N(A))$.

Computing $\text{Sol}(A, b)$: compute some solution s (Gauss-Jordan, Theorem 3.20) and a basis $B = \{v_1, v_2, \dots, v_{n-r}\}$ of $N(A)$ (Theorem 4.36). Then

$$\text{Sol}(A, b) = \left\{ s + \sum_{i=1}^{n-r} \lambda_i v_i : \lambda_i \in \mathbb{R} \text{ for } i \in [n-r] \right\}.$$